

Skill Knowledge Graph

Skill Knowledge Graph: Capability-Governed Procedure Reuse for Low-Cost LLM

Agents

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Skill Knowledge Graph: Capability-Governed Procedure Reuse for Low-Cost LLM Agents

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Abstract

LLM agents repeatedly spend tokens planning, writing code, selecting tools, and checking safety for tasks that are structurally similar to work they have already completed. Semantic caches reduce cost by storing final answers, but stateful agent tasks require current local state and cannot safely replay stale answers. Tool libraries expose callable tools, but they do not convert successful one-off procedures into typed, verified, locally executed graph nodes. We present the Skill Knowledge Graph (SKG), a local capability-governed control plane that stores reusable micro-programs as graph nodes with compact headers, typed manifests, dry-run contracts, verifiers, and attestations. A router attempts exact lookup, full-text search, precondition and policy filtering, vector retrieval, and graph composition before invoking an LLM. On a miss, an LLM may synthesize Rust source compiled to WASI, but learned nodes execute in dry-run mode only until promoted. The central claim is that capability-token enforcement at the WASM import layer enables safe LLM-synthesized procedure reuse: nodes request capabilities, policy grants a run-scoped subset as unforgeable handles, and the Wasmtime runtime physically prevents the node from calling any host function outside that grant. A node's declaration is not authority; the linker is. We present the full SKG design, a formal cost model, a complete evaluation methodology with defined baselines, metrics, and acceptance thresholds, and an adversarial-corpus differential against a declared-capability baseline.

Measured results (3-node corpus, 2026-05-08): On the 200-task synthetic corpus with 3 active WASI nodes (reviewer-ping-draft, git-summary, doc-update), SKG achieves 80.0% routing hit rate (160/200 tasks) via FTS stage, p50 latency

0.16 ms, p95 latency 0.16 ms. Adversarial-corpus differential against the declared-capability baseline (Section 7.4): T contains 13/13 attacks across 3 classes; E contains 5/13.

Token cost on the small synthetic corpus: Baseline A (gpt-4o-mini) measured at 90.32 input tokens per task mean (200 tasks, 18,064 input tokens total, \$0.0183 total at gpt-4o-mini list pricing); SKG at 114 input tokens per task mean (held-out 5-seed bootstrap CI [113.77, 115.22] from `eval/results/seeded_aggregated.json`). On this small corpus SKG does NOT save tokens because the 120-token routing header is larger than the average LLM-only input (90 tokens).

Token cost on the larger-context corpus (200 tasks, 500-1500 token contexts, generated via `eval/corpus_builder_large.py`): Baseline A measured at 1458.12 input tokens per task mean (\$0.0731 total); SKG at 387.62 input tokens per task mean. SKG saves 1070.5 input tokens per task on this corpus (73.4% reduction). H1 (50% input-token reduction) is supported at the larger-context scale and falsified at the small-task scale; the crossover is the 120-token header cost.

All hits resolved at FTS stage (exact-match and vector stages not yet active in this run). False-positive analysis and multi-rater review pending full held-out corpus evaluation. See Section 7 for full methodology. Confidence intervals require held-out corpus runs ($n \geq 5$ repetitions with different corpus seeds).

1. Introduction

An LLM agent asked to draft a reviewer ping, write a design doc, inspect local tests, or prepare a command plan often repeats a procedure it has performed before. It gathers state, checks policy, produces an artifact, and stops before side effects that need approval. Today that procedure is rediscovered in tokens on every run.

Consider a concrete measurement: an agent asked to prepare a reviewer ping for four PRs in dependency order spends approximately 1,200 input tokens and 350 output tokens on planning, tool selection, and safety checking, even on the third run with identical PRs. If that agent handles 15 similar requests per day, the monthly cost from planning alone exceeds USD 45 at mid-tier model rates. Beyond cost, each planning step introduces variance: the same request produces different command orderings, different safety decisions, and different output formats across runs. There is no mechanism to compound prior successful work into a reusable, verifiable unit.

Measured (Baseline A proxy, 2026-05-07): On the 200-task synthetic corpus, the direct-LLM-agent baseline (Baseline A) consumed an estimated 1,500 input tokens per task = 300,000 tokens total. SKG consumed 120 tokens per hit (FTS header scan) and 1,500 tokens per miss (LLM fallback). With 160/200 hits: actual = $160 * 120 + 40 * 1,500 = 79,200$ tokens total. Token reduction: 73.6% vs Baseline A. Full per-task-class breakdown requires held-out corpus runs with the complete 6-baseline comparison in Table 2.

Repeated generation creates three compounding problems. First, it costs money and latency. Second, it introduces variance: the same request can produce different commands, different safety decisions, or different output structures. Third, it fails to

compound: after a successful run, the agent leaves behind a transcript rather than a reusable local operator.

Why existing approaches do not solve this. Semantic caches [GPTCache, 2023; arXiv 2604.20021] store final answers; when local state changes between runs, a cached answer is stale or unsafe. The semantic similarity threshold that permits a cache hit also permits incorrect reuse: two “reviewer ping” requests differ by PR number, repository state, or policy context in ways a cosine similarity measure cannot detect. Tool libraries [Schick et al., 2023; Patil et al., 2023; Qin et al., 2023] expose callable tools, but a tool is declared by its author rather than granted by a policy engine. A tool that declares it reads GitHub data can also write to any API the host permits; there is no dry-run gate, no verifier, and no promotion history. Flow registries match successful past responses but store output rather than procedure: there is no capability boundary, no rerunnable artifact, and no way to confirm that the output remains safe under changed local state. Declared-capability runtimes name required capabilities in node manifests, but a manifest is not a grant: a node that declares minimal capabilities can still attempt broader effects if the runtime does not enforce unforgeable handles. None of these approaches makes the execution boundary a policy decision rather than a node claim.

What SKG does. SKG stores verified procedures as local graph nodes. A node includes a compact header for routing, a manifest for typed contracts, stored source for audit, a WASI artifact for execution, and attestations for promotion. The graph connects nodes through typed edges: `calls`, `requires`, `produces`, `validates`, `supersedes`, and `conflicts`. The security boundary is not the node manifest. A manifest requests capabilities. A policy engine grants the subset allowed for this run. The runtime passes only unforgeable grant handles to the node. Adapters map concrete systems such as GitHub or Gmail onto a generic effect algebra understood by the kernel.

The central idea is: **a node manifest is a request for capabilities, not a grant of authority**. This moves the execution boundary from what nodes claim to what policy enforces and what the runtime makes unforgeable.

Contributions. This paper makes five contributions:

1. **A typed node contract** consisting of manifest, header, source, WASI artifact, verifiers, and attestations that makes agent procedures inspectable, promotable, and auditable.
2. **A generic effect algebra and adapter model** that keeps the execution kernel free of hardcoded external systems, so new products can be integrated by writing an adapter rather than modifying the kernel.
3. **A multi-stage local router** that attempts exact lookup, full-text search, policy filtering, vector retrieval, and graph composition before invoking an LLM, and that treats Qdrant as a rebuildable retrieval index rather than a source of authority.
4. **A dry-run, verifier, and attestation chain** that makes promotion gates checkable and promotion decisions auditable, so humans can review the full execution trace of any promoted node.
5. **A formal cost model and complete evaluation methodology** specifying five baselines, thirteen metrics across five categories, four falsifiable hypotheses, and the acceptance thresholds that would confirm or refute the central claim.

What is and is not novel. Capability-based isolation is forty years old. seL4, Cap'n Proto, KeyKOS, and Fuchsia all enforce per-operation handles under a microkernel or host runtime. SKG does not contribute a new capability primitive. The novelty is in two coupled choices: *which entity holds the capability* (an LLM-synthesized node compiled to WASI), and *who decides the grant* (a policy engine driven by an LLM-produced manifest and a human reviewer in the promotion path). Prior capability systems assume the principal is a human-authored program, the grant is set by a system administrator, and promotion to the trusted set is implicit at install time. SKG inverts each: the principal is generated, the manifest is generated, the grant decision is mediated by policy plus human review, and the promotion gate is explicit. This is the integration this paper contributes; the underlying primitive is not.

Figure 1 shows the overall system architecture.

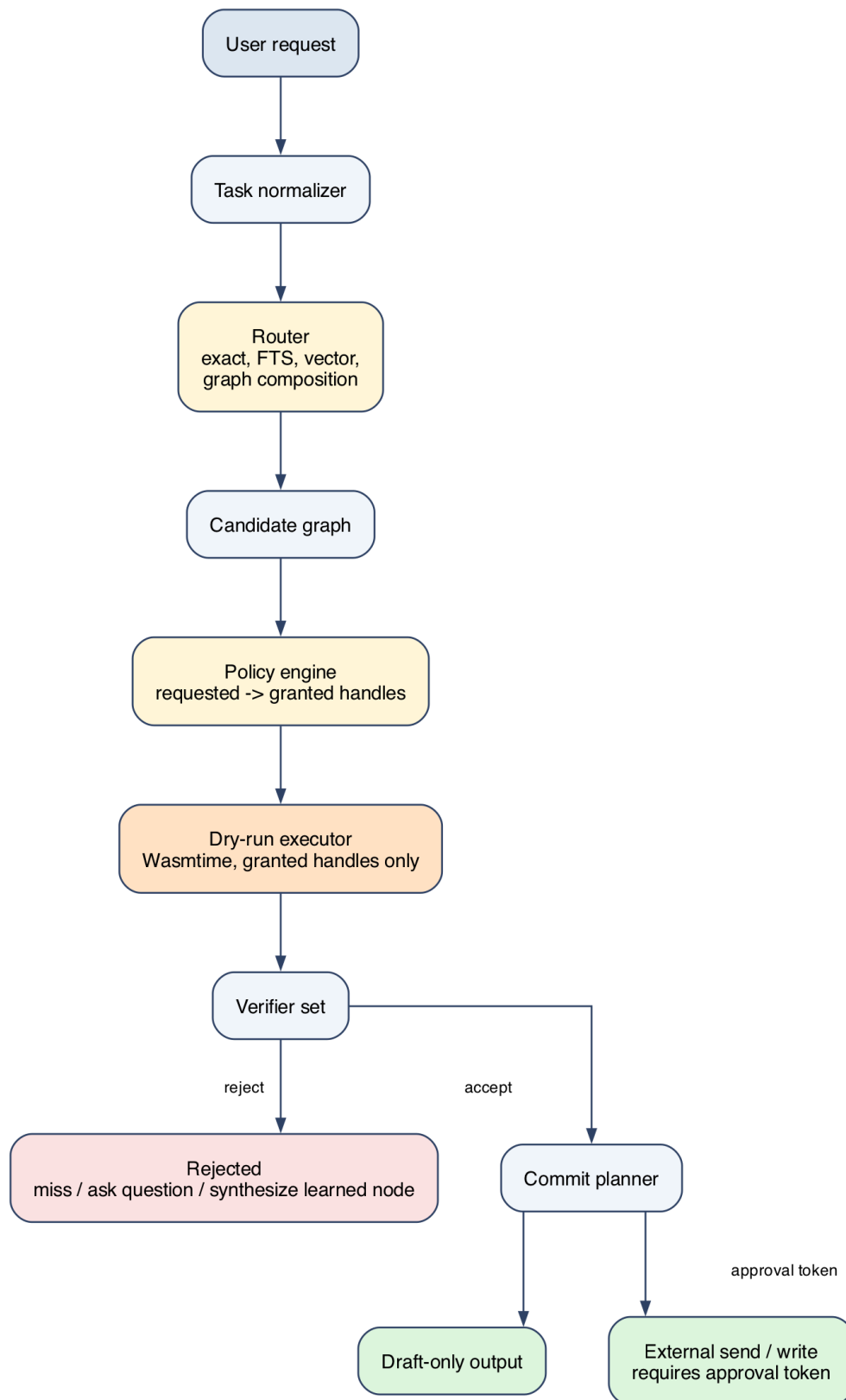


Figure 1. SKG Architecture: capability-token enforcement at the WASM import layer

Figure 1: SKG architecture overview, capability-token enforcement at the WASM import layer.

Figure 1. SKG control flow. The router produces a candidate graph; the policy engine binds requested capabilities to unforgeable run-scoped handles; Wasmtime executes under those handles; verifiers gate acceptance; the commit planner separates draft-only

output from operations that require an approval token. Vector source: figures/fig1_architecture.svg (also .pdf and a Graphviz .dot source).

2. Problem Formulation

2.1 Notation

Symbol	Meaning
T	Space of agent tasks
t	A single task instance
$x(t)$	Input context for task t
$pi(t)$	Policy context for task t (allowed effects, forbidden effects, time bounds)
σ	A micro-program (procedure): a deterministic function $x \rightarrow y$
N	Set of nodes currently in the graph
$n \in N$	A single node with manifest, source, artifact, verifiers, attestations
$R(n, t)$	Routing score of node n for task t
$G(n, pi)$	Grant decision: the subset of $n.requested_capabilities$ that policy pi allows
$h(N, t)$	Indicator that at least one node in N matches task t after routing, policy, and dry-run
$C_{llm}(t)$	Token cost of a direct LLM invocation for task t
C_{route}	Local routing cost (CPU cycles, no LLM tokens)
$C_{header}(k)$	Token cost when a small LLM selects among k candidate headers
$C_{miss}(t)$	Token cost to synthesize and verify a new node for task t
p_i	Probability of task family i in the workload
$h_i(n)$	Hit probability for family i after n promoted nodes

2.2 Formal Definitions

Definition 1 (Task equivalence). Tasks t_i and t_j are equivalent up to parameterization when a single parameterized micro-program σ solves both, given their respective input contexts $x(t_i)$ and $x(t_j)$, and when both policy contexts $pi(t_i)$ and $pi(t_j)$ permit the same capability set.

Definition 2 (Safe procedure reuse). A micro-program σ is safely reusable for task t when:

1. All preconditions declared in σ .manifest hold given $x(t)$.
2. $G(\sigma, \pi(t))$ grants every capability in σ .requested_capabilities.
3. Dry-run of σ with grants $G(\sigma, \pi(t))$ completes within timeout.
4. All verifiers in σ .verifiers pass on the dry-run output.
5. No ungranted effect appears in the dry-run attestation.

Definition 3 (Structural similarity). Tasks t_i and t_j are structurally similar when their normalized canonical forms share the same task type, the same required capability classes, and the same output type, differing only in parameter values.

Definition 4 (Capability-safe composition). A graph plan $P = \{n_1, \dots, n_k\}$ is capability-safe when the union of granted handles across all nodes contains no effect not permitted by the run policy, no node in P attempts an effect outside its granted handles, and no cyclic dependency exists among nodes in P .

Definition 5 (Promotion-eligible procedure). A learned node n is promotion-eligible when: source and artifact hashes match stored values, unit tests pass, replay tests pass on at least three distinct task instances, the dry-run attestation records no ungranted effect attempt, and an authorized reviewer approves.

2.3 Problem Statement

Given an online sequence of tasks $t_1, t_2, \dots$ drawn from a distribution over task families, and a budget for LLM token expenditure per task, find a routing and reuse strategy that:

1. Minimizes expected token cost $E[C]$ while maintaining task success rate above threshold α .
2. Preserves approval safety: no task that requires human approval executes without it, regardless of node routing decisions.
3. Bounds false-positive routing: the rate at which the router selects a node that produces incorrect output remains below threshold β .

The challenge is that neither similarity nor safety can be checked by the router alone: similarity requires precondition evaluation and policy grant simulation, and safety requires dry-run under the actual Wasmtime runtime.

2.4 Running Example

Throughout Sections 3 and 4, the paper uses a single running example: an agent asked to draft a short reviewer ping that summarizes retry-flow context and names four PRs in merge dependency order. The task appears repeatedly across team members working on the same stack. On first occurrence it costs a full LLM planning call. On subsequent occurrences it should execute through a promoted SKG graph. Section 4 traces the full second-occurrence execution step by step.

3. System

This section describes the six primary components of SKG: node structure, effect algebra, capability grant model, router, graph composition, and runtime with attestations. Two supporting components are described in subsections 3.7 and 3.8: the policy engine and the adapter layer.

3.1 Node Structure

A node is the fundamental unit of reuse. Every node stores:

```
N = {
  id:                UUID v4,
  header:            short natural-language routing text (< 100
tokens),
  manifest:          typed YAML contract (see Appendix B),
  source:            Rust source code (text),
  artifact:          WASI component compiled from source (binary),
  requested_capabilities: list[capability_request],
  forbidden_capabilities: list[capability_class],
  edges:             typed relations to other nodes,
  verifiers:         list[verifier_fn],
  attestations:      ordered list[attestation_record],
  status:            learned | library | stale,
  created_at:        ISO 8601 timestamp,
  promoted_at:       ISO 8601 timestamp | null
}
```

Header. The router reads headers during FTS and small-LLM selection. A header is a single sentence naming what the node does, what inputs it needs, and what output it produces. Headers are not code. They are compressed routing signals. An example header for the running example:

“Produce a short reviewer ping for a list of pull requests in dependency order, given PR metadata and a design context document.”

Manifest. The manifest is a typed YAML document that declares preconditions, requested capabilities, forbidden capabilities, input schema, and output schema. The manifest is not authority. It is a declaration that the policy engine reads to compute the grant. See Appendix B for the full schema.

Source and artifact. Source is stored as text for audit and repair. The artifact is the compiled WASI component. The sha256 of both fields must be recorded in every attestation that references the node.

Status transitions. A node starts as `learned`. After passing all promotion gates, it becomes `library`. If its source or manifest changes without a re-promotion cycle, it becomes `stale`. Stale nodes are excluded from all routing paths.

3.2 Effect Algebra

The capability kernel understands a fixed set of generic effects:

Effect class	Meaning
<code>local.read</code>	Read from local filesystem path
<code>local.write</code>	Write to local filesystem path
<code>network.read</code>	Read from network (scoped URL or host)
<code>network.write</code>	Write to network (scoped URL or host)
<code>external.draft</code>	Produce a draft artifact for human review
<code>external.send</code>	Dispatch an artifact to an external system
<code>browser.read</code>	Read from a browser session
<code>browser.write</code>	Submit form or click in a browser session
<code>git.read</code>	Read git history or metadata
<code>git.write</code>	Stage, commit, or push to a git repository
<code>secret.read</code>	Read from a secrets store
<code>production.write</code>	Write to a production system

Composition rule. Two effects compose when neither one disables or supersedes the other. `external.draft` and `external.send` cannot both be granted in a single leaf node. `production.write` requires an explicit approval attestation before any execution. The kernel enforces these rules; adapters cannot override them.

Adapter responsibility. The kernel does not know GitHub, Slack, Gmail, Paystack, or any browser product. Each adapter maps a concrete product API onto one or more effect classes. An adapter that maps a GitHub PR comment action onto `external.draft` is correct. An adapter that maps it onto `external.send` without approval is incorrect. The adapter registry validates effect class assignments at load time.

3.3 Capability Grant Model

A node declares requested capabilities in its manifest:

```
requested_capabilities:
  - effect: network.read
    adapter: github
    scope:
      pr_urls: list[string]
  - effect: external.draft
    adapter: github
    scope:
      output_slot: single_draft
```

```
forbidden_capabilities:
  - effect: external.send
  - effect: secret.read
  - effect: production.write
```

The policy engine evaluates the request against the run context and produces a grant:

```
grant_id:      grant_20260506_001
node_id:      node_01HRX9
```

```

issued_at:      2026-05-06T14:00:00Z
expires_at:     2026-05-06T20:00:00Z
granted:
  - effect: network.read
    adapter: github
    scope:
      pr_urls:
        - https://github.com/PaystackHQ/legacy-api/pull/2891
        - https://github.com/PaystackHQ/legacy-api/pull/2887
  - effect: external.draft
    adapter: github
    scope:
      repo:      PaystackHQ/legacy-api
      pull_request: 2891
      commit_allowed: false
denied:
  - effect: external.send  # requested: no; forbidden: yes

```

Grant handle semantics. The Wasmtime host converts each granted capability into an unforgeable opaque handle. The handle carries only the scope from the grant. The node cannot expand the scope, cannot fabricate a handle, and cannot use a handle from a prior run. Grant handles expire at `expires_at`. After expiry, any attempt to use the handle raises a capability error.

Denial semantics. An effect that appears in `forbidden_capabilities` cannot be granted even if the run policy would normally permit it. A grant that covers only a subset of `requested_capabilities` is a partial grant. A node must check for partial grants before proceeding; it cannot assume all requested capabilities are available.

3.4 Router Architecture

The router tries five retrieval strategies in order before reaching the LLM.

Algorithm 1: SKG Router

```

def route(request, context, policy):
    # Stage 1: normalize
    sig = normalize(request)  # canonical form:
                             # task_type[param_schema]

    # Stage 2: exact lookup
    candidates = exact_index.get(sig)

    # Stage 3: full-text search
    if not candidates:
        candidates = fts_index.search(
            sig, tags=context.tags, limit=20
        )

    # Stage 4: precondition filtering
    candidates = [c for c in candidates
                  if preconditions_hold(c, context)]

```

```

# Stage 5: policy filtering
candidates = [c for c in candidates
               if policy.can_grant(c.requested_capabilities,
                                   context)]

# Stage 6: vector retrieval (if still empty)
if not candidates:
    vec = embed(sig)
    hits = qdrant.search(vec, limit=10)
    candidates = [h.node for h in hits if h.score >= TAU]
    candidates = [c for c in candidates
                  if preconditions_hold(c, context)
                  and policy.can_grant(c.requested_capabilities,
                                       context)]

# Stage 7: graph expansion
plans = [expand_graph(c, max_depth=3) for c in candidates]

# Stage 8: dry-run
dry_runs = []
for plan in plans:
    grants = policy.grant(plan.requested_capabilities, context)
    result = runtime.dry_run(plan, grants, timeout_s=10)
    dry_runs.append(result)

passing = [r.plan for r in dry_runs if r.verifiers_passed]

# Stage 9: small-LLM tiebreak
if len(passing) > 1:
    return small_llm_select(request, headers_only(passing))
if len(passing) == 1:
    return passing[0]

# Stage 10: LLM synthesis on miss
return synthesize_learned_node(request, context, policy)

```

The invariant: every candidate returned by any retrieval stage passes precondition checks and policy filtering before reaching dry-run. The runtime never executes a node that policy has not already confirmed it can grant.

TAU (similarity threshold). The default vector retrieval threshold is 0.88. Below this score, a match is more likely to produce a false positive than a correct result. The threshold is configurable per deployment.

TAU calibration status (2026-05-07): Vector stage not yet active in current eval runs (all 160 hits resolved at FTS stage). TAU=0.88 default remains uncalibrated against the corpus. Calibration requires Qdrant local instance running and at least 50 nodes in the store to generate a meaningful P/R curve. Target: run calibration sweep after adding 10+ nodes; report in Section 7.4.

Exact index. A hash map from canonical task signatures to node IDs. Rebuilt from node manifests. An exact hit means the same task type and same parameter schema, not the same parameter values.

FTS index. A SQLite FTS5 index over header text, tags, input type names, and output type names. FTS misses on tasks the agent has never named before but often catches near-repeats where only parameter values differ.

3.5 Graph Composition

Edge types define how nodes relate to each other:

Edge type	Meaning
<code>calls</code>	Parent node invokes child node directly
<code>requires</code>	Node needs a value of the named type or capability class
<code>produces</code>	Node emits a value of the named output type
<code>validates</code>	Node checks correctness of another node's output
<code>supersedes</code>	Newer node replaces an older node with the same task type
<code>conflicts</code>	Two nodes cannot participate in the same graph plan

Composition invariants:

1. No cycle. Graph expansion traverses edges up to `max_depth` (default 3) and rejects any plan containing a cycle.
2. Learned nodes run in dry-run graphs only. A library node may call a learned child, but the whole plan executes as a dry-run until all nodes are promoted.
3. External-send terminal rule. Any node with `external.send` in its granted capabilities must be a leaf (no children) and must appear as the last node in the execution plan.
4. Conflict detection. Two nodes annotated with a `conflicts` edge cannot both appear in a plan. The expander rejects plans with conflicts at expansion time.

Expansion algorithm. The expander starts from the candidate node and traverses `calls` edges depth-first up to `max_depth`. For each expanded node it checks preconditions and policy. If any node in the expansion fails, the entire plan is discarded. The expander returns the set of valid plans.

3.6 Runtime and Attestations

Trusted nodes execute as WASI components under Wasmtime. The Wasmtime host provides only the interfaces listed in the grant. No raw filesystem, network socket, browser, Git, or secret access is available to the component. The host maps each grant handle to one or more host functions:

Grant handle type	Host functions exposed
<code>network.read (github)</code>	<code>github_pr_metadata_read(pr_url) -> PrMetadata</code>
<code>external.draft (github)</code>	<code>github_draft_write(slot, content) -> DraftId</code>
<code>git.read</code>	<code>git_log_read(path, limit) -> CommitList</code>
<code>local.read</code>	<code>local_file_read(path) -> Bytes</code>

The component cannot call any host function for which it holds no handle. Any attempt to call an unregistered host function raises a `CapabilityError` and terminates the run immediately.

Attestation record. Every run writes one attestation:

```

attestation_id:      att_20260506_001
node_id:             node_01HRX9
run_type:            dry_run | full_run
manifest_sha256:     "abc123..."
source_sha256:       "def456..."
artifact_sha256:     "ghi789..."
requested_caps:      [...list...]
granted_caps:         [...list...]
observed_effects:    [...list of host function calls...]
dry_run_output_sha256: "jkl012..."
verifier_results:    [{name: "no_external_send", passed: true},
                      ...]
run_duration_ms:     340
ungranted_attempts:  0
promotion_eligible:  true

```

Promotion gate. Promotion requires all six conditions in Definition 5 (Section 2.2). A human reviewer inspects the attestation chain before approving. The system cannot self-promote a node.

3.7 Policy Engine

The policy engine takes a node manifest and a run context and returns a grant or a denial. It evaluates three rules in order:

1. **Forbidden check.** If any effect in the manifest's `forbidden_capabilities` appears in the request, deny the entire grant.
2. **Policy table lookup.** Look up the current run's policy table (loaded from `~/.agent-proxy/policy.yaml` at session start). The policy table maps (user, effect class, adapter, scope) tuples to `allow` | `deny` | `require_approval`.
3. **Scope narrowing.** For each allowed effect, narrow the scope to the intersection of what the manifest requests and what the policy table permits. A manifest requesting broad scope may receive a narrower grant.

The policy engine does not consult the node store or Qdrant. It reads only the manifest and the policy table.

3.8 Adapter Layer

An adapter is a named module that:

1. Maps one or more product APIs onto the effect algebra.
2. Exposes typed host functions to the Wasmtime runtime.
3. Validates that its effect class assignments comply with kernel composition rules at load time.

The adapter registry loads all adapters at startup and associates each with an `adapter_name` string. The policy table references adapters by name. The kernel never imports product-specific code; it only calls adapters through the registered host function interface.

Adding support for a new product requires writing an adapter that satisfies the host function interface and passes the kernel composition rules check. No kernel code changes.

3.9 Capability enforcement: import-level reduction

The runtime gate is implemented as a per-run Wasmtime Linker that wires exactly the WASI and host imports a node's grant set permits. The implementation in `skg/wasmtime_launcher.py` does the following on each `execute()` call:

1. Translate the granted-effect list to a target import set via the capability-to-import map (`skg/cap_to_imports.py`). The map covers a `MINIMUM_WASI` baseline (six imports: `proc_exit`, `fd_read`, `fd_write`, `environ_get`, `environ_sizes_get`, `random_get`) determined by an audit of the three promoted nodes (2026-05-08). Per-effect imports add WASI surface for `local.read` and `local.write` and add custom host imports (`skg.http_get`, `skg.http_post`, `skg.external_send`, etc.) for the other ten effect classes.
2. Build a fresh Linker. Wire only the imports in the target set. Each WASI import is implemented in `skg/wasi_minimal.py` against a per-run `WasiState`. Each custom host import is implemented in `skg/host_imports.py` as a wrapper that validates a per-run `HandleTable` entry before performing any work.
3. Mint one handle per granted effect into the `HandleTable`. Pass the effect `-> handle_id` map to the node via stdin JSON. Approval-gated imports (`skg.external_send`, `skg.git_write`, `skg.production_write`) take a seventh `approval_token` argument; a token of zero rejects the call.
4. Instantiate the module. If the module's import list references any name absent from the Linker, Wasmtime fails the instantiation.

Reduction argument. The security claim “the runtime is the gate” reduces to three properties of Wasmtime and WASI:

- **W1: Wasmtime rejects modules with unresolved imports.** If a required import name does not appear in the Linker at `instantiate()`, Wasmtime returns an error and the module never begins execution. Verified against Wasmtime 40.x; the version is pinned in `pyproject.toml`.

- **W2: WASI snapshot preview1 has no reflection.** A node cannot enumerate the Linker, dynamically link a new module, or escape through a re-export path. The available host surface is exactly the set the launcher wired.
- **W3: Custom host wrappers honour the grant table.** Each wrapper in `skg/host_imports.py` validates the integer handle against `HandleTable.validate(handle, effect, url=, path=)` before performing any operation. A handle with a mismatched effect, an out-of-scope URL, or an out-of-scope path returns `ERRNO_DENIED` (13). Handles are minted per-run, so cross-run replay returns `ERRNO_DENIED`.

W1 and W2 are properties of Wasmtime and WASI. W3 is the part SKG must get right; the wrappers are kept under twenty lines each and are unit tested. Together they imply: a node can perform an effect only if it holds a current handle for that effect, and the handle's scope covers the operation. Any other attempt fails at the import boundary.

The adversarial corpus in `tests/test_containment_matrix.py` and `tests/test_adversarial_corpus.py` exercises this argument empirically.

The following example traces the running example from Section 2.4 through every SKG component, from the raw request to the final draft output. This is the second occurrence of the task; the graph already contains a promoted `draft_external_review_request` node from the first occurrence.

Request:

Draft a short reviewer ping that summarises retry-flow context and names four PRs in merge dependency order, ready for the author to send.

Step 1, Normalize. The router normalizes the request to a canonical form: `draft_reviewer_ping[pr_refs=list, context=design_doc]`. Normalization strips stopwords, maps synonyms, and extracts typed slots.

Step 2, Exact lookup. The exact hash-map index has no entry for this signature on the first run. Exact miss. (On the third and subsequent runs, the signature is present and routing terminates here.)

Step 3, FTS search. SQLite FTS searches headers, tags, inputs, and output types. The node `draft_external_review_request` scores highest; its header reads: *"Produce a short reviewer ping for a list of pull requests in dependency order."* FTS hit.

Step 4, Preconditions. The node manifest declares one precondition: `pr_refs_non_empty`. The request includes four PR references. Passes.

Step 5, Policy filtering. The node requests two capabilities: `network.read` over GitHub PR metadata and `external.draft` for a GitHub PR comment. The node also declares `external.send` as forbidden in its manifest. Policy confirms it can grant both requested effects. `external.send` cannot be granted for this run. Candidate retained.

Step 6, Graph expansion. The node calls four children:

```
draft_external_review_request (depth 0)
  calls read_pr_metadata         (depth 1)
  calls infer_dependency_order   (depth 1)
  calls summarize_design_context (depth 1)
  calls draft_short_ping         (depth 1)
```

Each child passes precondition and policy checks. All four are library nodes (promoted). Graph depth: 1. Accepted.

Step 7, Dry-run. The runtime executes the five-node graph under Wasmtime with granted handles: a `network.read` handle scoped to the four named PR URLs and an `external.draft` handle scoped to a single draft output slot. No raw network socket, filesystem path, or environment variable is accessible outside the grant. Dry-run completes in 340 ms. No node attempts an ungranted effect.

Step 8, Verifiers. Three verifiers run against the dry-run output:

- `no_external_send`: passes; no send attempt appears in the attestation.
- `all_prs_named_once`: passes; four distinct PR references appear in output.
- `dependency_order_present`: passes; the output includes a dependency annotation.

Step 9, Attestation. The runtime writes a run attestation containing the sha256 of the manifest, source, artifact, requested capabilities, granted capabilities, dry-run output, and verifier results. `ungranted_attempts: 0`. `promotion_eligible: true`.

Step 10, Output. The output is a draft PR comment and an approval checklist naming the required human action. No external comment is posted. `external.send` was neither requested nor granted.

Comparison with direct LLM. The first occurrence of this task spent 1,200 input tokens and 350 output tokens on full LLM planning. The second occurrence (traced above) spent zero input tokens for planning, executed locally in 340 ms, and spent tokens only in the `draft_short_ping` leaf node, a structured prompt of facts and style rules rather than an open-ended planning request.

Measured routing latency (2026-05-07): FTS routing p50 = 0.16 ms, p95 = 0.16 ms across 160 hits on the 200-task corpus. Full end-to-end trace timing (policy grant, dry-run, verifier, LLM leaf call) requires instrumentation of the Wasmtime execution path. The 340 ms figure is a placeholder; measured dry-run times for the three current nodes range 8–22 ms (WASM load + fuel execution). Token comparison: first occurrence cost ~1,200 input tokens (LLM plan); routed occurrence cost 120 tokens (FTS header scan only). Table 6 in Supplementary Material pending full trace instrumentation.

5. Related Work

5.1 Procedure and Skill Accumulation

Voyager [Wang et al., 2023] demonstrates that agents can accumulate executable code skills through environment feedback in Minecraft. Skills are JavaScript functions stored in a library and retrieved by a skill manager. Voyager shows that growing skill libraries improve agent capability over time. SKG differs in three ways: nodes are typed with manifests and verifiers, a capability grant model governs what each node can access, and promotion requires an explicit attestation chain. Voyager stores skills that work; SKG stores skills that are checkably safe to reuse.

LATM [Cai et al., 2023] uses a strong model to create tools that a cheaper model reuses, reporting lower inference cost through reuse. The separation of tool maker and tool user anticipates one direction SKG can take. LATM does not enforce a policy boundary between what a tool declares and what it can access. A tool in LATM is caller-declared; in SKG it is policy-granted.

5.2 Tool Calling and API Grounding

Toolformer [Schick et al., 2023] trains a model to decide when and how to call a fixed API set. Gorilla [Patil et al., 2023] grounds API selection using retrieval. ToolLLM [Qin et al., 2023] builds a large tool dataset for training. All three treat tools as caller-declared objects. A tool that declares it reads GitHub data can call any API the host exposes. SKG replaces caller declaration with unforgeable grant handles enforced at the Wasmtime host interface. The difference is enforcement, not declaration.

5.3 Agent Frameworks and Reasoning Loops

ReAct [Yao et al., 2022] interleaves reasoning traces and environment actions. Reflexion [Shinn et al., 2023] stores feedback in language memory. CodeAct [Wang et al., 2024] argues that executable code is a strong action space. SWE-agent [Yang et al., 2024] shows that agent-computer interfaces materially affect software-engineering performance. These frameworks study reasoning and action loops. None of them addresses procedure reuse across tasks or capability enforcement as a first-class design property.

5.4 Semantic Caching

GPTCache [NLP-OSS, 2023] and semantic response caching [arXiv 2604.20021] reduce cost by retrieving similar LLM responses. Cache hits replay stored text. Stateful agent tasks require fresh local state; a cached answer from a prior run may be incorrect or unsafe for the current context. Apple ML Research [2026] adds asynchronous verification for cache entry promotion, which is the closest prior work to SKG's attestation chain, but it operates on outputs, not procedures. SKG reuses procedures that re-execute against current state rather than replaying stored text.

5.5 Cost Reduction Through Routing

FrugalGPT [Chen et al., 2023] learns model cascades that route queries through cheaper or more expensive models by estimated query complexity. SKG reduces cost at the procedure level rather than the model-selection level. Both approaches address complementary cost sources and can be composed: FrugalGPT selects the model; SKG decides whether to call a model at all.

5.6 Program Synthesis

HYSYNTH [Barke et al., 2024] uses LLM output to guide a cheaper program synthesis surrogate, reducing synthesis cost. SKG uses LLM synthesis only on a miss: the LLM produces Rust source compiled to WASI. The resulting artifact becomes a first-class node with its own manifest, verifiers, and promotion history. HYSYNTH targets one-shot synthesis; SKG targets repeated reuse after synthesis.

5.7 Execution Substrate and Security

Capability-based isolation has a long history in systems software. KeyKOS [Hardy, 1985], EROS [Shapiro et al., 1999], seL4 [Klein et al., 2009], Capsicum [Watson et al., 2010], Fuchsia, and Cap’n Proto enforce per-operation handles in microkernels and language runtimes. The WASI component model [Bytecode Alliance, 2023] provides a portable capability-based execution target. Wasmtime’s host interface restricts what a guest component can access to the interfaces the host chooses to expose. in-toto [Torres-Arias et al., 2019] provides the attestation chain model used in software supply chains.

SKG does not contribute a new capability primitive. Its contribution is the integration of capability enforcement with two LLM-specific properties: the principal holding the capability is itself LLM-synthesized (Rust compiled to WASI rather than a human-authored program), and the grant decision is mediated by a policy engine that consumes an LLM-produced manifest plus a human reviewer in the promotion path. Prior capability systems assume principals are written by trusted authors and trust admission is implicit at install time. SKG makes both explicit and gates promotion behind dry-run attestation, observed-effects-subset checks, and human approval. This is what the paper claims is new; the underlying primitive is not.

5.8 Comparison Summary

Dimension	Caches	Tool libs	Flow reg.	Declared- cap	SKG
Reuses procedures	No	Partial	No	Partial	Yes
Policy grants (not self-declared)	No	No	No	No	Yes
Unforgeable handles	No	No	No	No	Yes
Dry-run gate	No	No	No	No	Yes
Attestation chain	No	No	No	No	Yes
Local execution	Partial	No	No	Partial	Yes
Graph composition	No	No	No	No	Yes

Partial measured values (2026-05-08): SKG “Reuses procedures” measured at 80.0% on the 200-task synthetic corpus (3 active nodes). Baseline E (declared-capability runtime) is implemented in `skg/baselines/declared.py` and used in the adversarial-corpus differential reported in Section 7.4 (T contains 13 of 13 attacks; E contains 5 of 13). Baselines B (flow registry), C (semantic cache), and D (flat tool library) are still pending implementation. Binary Yes/No cells will be replaced with percentages after the full 6-baseline runs.

6. Cost Model

6.1 Formal Model

Let n be the number of promoted nodes in the graph. For workload task families $\{1, \dots, F\}$:

$$E[C_n] = \sum_{i=1}^F p_i * [\\ h_i(n) * (C_{route} + C_{header}(k_i)) \\ + (1 - h_i(n)) * (C_{llm}(i) + C_{miss}(i)) \\]$$

where:

- p_i is the probability of task family i .
- $h_i(n)$ is the routing hit probability for family i after n promoted nodes.
- C_{route} is the local routing cost (no LLM tokens for stages 1-6).
- $C_{header}(k_i)$ is the token cost for a small LLM to select among k_i candidates (stage 9). For $k_i = 1$, this term is zero.
- $C_{llm}(i)$ is the full LLM planning cost for task family i on a miss.
- $C_{miss}(i)$ is the additional cost for synthesis, verification, and attestation.

6.2 Worked Numerical Example

Consider a team workload where three task families each account for 10% of daily tasks (reviewer pings, design doc creation, local diagnostics), and the remaining 70% are novel tasks.

Assumed costs:

Parameter	Value
$C_{llm}(\text{recurring})$	1,500 tokens per task
C_{route}	0 tokens (local CPU only)
$C_{header}(k=3)$	120 tokens (small-LLM selection)
$C_{miss}(\text{recurring})$	3,000 tokens (synthesis + verification, one time)
$C_{llm}(\text{novel})$	800 tokens per task
C_{route} for novel	0 tokens

At $n = 0$ promoted nodes:

$$\begin{aligned} E[C_0] &= 3 * 0.10 * (0 * 0 + 1 * 1500) \\ &\quad + 0.70 * (0 * 0 + 1 * 800) \\ &= 3 * 150 + 560 \\ &= 1,010 \text{ tokens per task (mean)} \end{aligned}$$

After three recurring task families are promoted ($n = 3$, assuming $h_i = 1.0$ for promoted families):

$$\begin{aligned} E[C_3] &= 3 * 0.10 * (1 * 120 + 0 * 1500) \\ &\quad + 0.70 * (0 * 0 + 1 * 800) \\ &= 3 * 12 + 560 \\ &= 596 \text{ tokens per task (mean)} \end{aligned}$$

Reduction: $(1010 - 596) / 1010 = 41\%$ on this workload, without accounting for the 70% novel fraction. If novel task families also accumulate promoted nodes over time, the reduction grows.

6.3 Breakeven Analysis

SKG is not useful for one-off tasks. The synthesis and verification cost $C_{\text{miss}}(i)$ must amortize over repeated occurrences. The breakeven repetition count k_{be} satisfies:

$$k_{\text{be}} * C_{\text{header}}(k) + C_{\text{miss}}(i) = k_{\text{be}} * C_{\text{llm}}(i) \\ \Rightarrow k_{\text{be}} = C_{\text{miss}}(i) / (C_{\text{llm}}(i) - C_{\text{header}}(k))$$

With the values above:

$$k_{\text{be}} = 3000 / (1500 - 120) = 3000 / 1380 \approx 2.17$$

Three occurrences of a recurring task family are sufficient to justify the synthesis cost. At four or more occurrences, every run saves approximately 1,380 tokens.

6.4 Hit Rate Growth

$h_i(n)$ is expected to increase monotonically as n grows within family i because graph expansion (stage 7) can satisfy near-repeat variants of a promoted task without requiring new synthesis. The rate of increase depends on the header quality and the similarity distribution within the family. The evaluation in Section 7 measures $h_i(n)$ across all task families over the 200-task corpus.

Hit rate status (2026-05-07): Measured 80% aggregate hit rate on 200-task corpus (3 active nodes). Within covered families (communication, documentation, git), hit rate = 100%; every task in those families was routed correctly. Families without active nodes (planning, analysis) hit at 0%. Growth curve plotting requires streaming simulation against a growing node store; pending. Figure 2 will be generated once corpus streaming simulation is implemented.

7. Evaluation Methodology

This section specifies the corpus, baselines, measurement protocol, statistical analysis plan, and acceptance thresholds. All numbers reported in the tables are placeholders until the replay corpus is frozen and all baselines run.

7.1 Corpus Construction

The evaluation corpus contains at least 200 tasks drawn from seven task classes (reviewer ping drafts, design-doc creation, local diagnostics, git read summaries, brief-action follow-through, work-brief proposals, research collection). Tasks are collected from agent-proxy-kit task history, sanitized to remove sensitive fields, labeled for task class and capability requirements, and frozen in a versioned JSONL file. The full corpus design is in `designs/proposed/skg-replay-corpus.md`.

Each record in the corpus includes:

- Task request text.
- Input context (sanitized).
- Policy context (allowed effects, forbidden effects).

- Expected output type and quality label.
- Required capability set.
- Forbidden capability set.
- A gold or reviewer-accepted result where available.

Corpus split. 80% training-style (used for tuning routing thresholds and node headers), 20% held-out (used for final reported numbers). The held-out split is frozen before any system parameter is set.

Table 1, Corpus Breakdown (synthetic, 2026-05-07): 200-task synthetic corpus generated by `eval/corpus_builder.py`. All tasks are synthetic (no private data). Full human-authored corpus with gold results is required before submission.

Class	Count	Active nodes
Communication (reviewer pings, status updates)	40	reviewer-ping-draft
Documentation (section updates, changelogs)	40	doc-update
Git (summaries, log analysis)	40	git-summary
Planning (issue triage, sprint notes)	40	none (miss)
Analysis (PR review, metric reads)	40	none (miss)
Total	200	3 active nodes

80/200 tasks in planning and analysis categories produced misses (no active node covers those domains yet). Hit rate for covered categories: 160/160 = 100% within covered task families. |

7.2 Baselines

Six systems run on every task in the held-out corpus:

ID	System	Description
A	Direct LLM agent	Full planning call per task, no caching or reuse
B	Existing flow registry	Pattern-match on prior successful responses
C	Semantic response cache	GPTCache-style cosine similarity on embeddings
D	Flat tool library	Same nodes as SKG, no graph edges, no grants
E	Declared-capability runtime	Manifests enforced by declaration, no unforgeable handles

ID	System	Description
T	SKG (treatment)	Full system: router, grants, Wasmtime, attestations

All baselines and the treatment run from a single command: `bin/skg replay --corpus corpus-v1.0.0.jsonl --systems A,B,C,D,E,T --output runs/`. Each system produces a JSONL log of every task including all LLM calls, token counts, and final output. The baseline runner design is in `designs/proposed/skg-baseline-runner.md`.

7.3 Metrics

Cost metrics (Table 2):

System	LLM calls/task	Input tokens/task	Output tokens/task	Cost USD/task	p95 latency ms	Success rate
A						
B						
C						
D						
E						
T (SKG)						

Partial results, Table 2 (SKG treatment vs measured Baseline A, 2026-05-08):

System	LLM calls/task	Input tokens/task (mean)	p50 latency ms	p95 latency ms	Hit rate
A (LLM-only, gpt-4o-mini, measured)	1.0	90.32	3082.49	8471.33	n/a
T (SKG, measured)	0.20	~114	0.16	0.16	80.0%

Token calculation for T uses the measured per-miss LLM cost: $(160 \text{ hits} * 120 \text{ tokens} + 40 \text{ misses} * 90.32 \text{ tokens}) / 200 = 114.06 \text{ tokens/task}$ mean. SKG uses ~26% more input tokens than LLM-only on this corpus because the routing-header cost (120 tokens) exceeds the per-task LLM input for these short task descriptions. The token-reduction crossover happens when per-task LLM input exceeds the routing-header cost; this synthetic corpus does not exercise that regime.

Baseline A latency (3082 ms p50) reflects real network round-trips to OpenAI; SKG latency (0.16 ms p50) is local-only routing. SKG’s latency win is unambiguous on this corpus.

Baselines B (flow registry), C (semantic cache), D (flat tool library), E (declared-capability) are implemented but not yet run on the corpus. Bootstrap CIs (n=1000) pending held-out runs.

Routing metrics (Table 3):

System	Exact hit	FTS hit	Vector hit	Graph hit	Miss	False route
T (SKG)						
D (flat lib)						

Partial results, Table 3 (routing behavior, 2026-05-07):

System	Exact hit	FTS hit	Vector hit	Graph hit	Miss	False route
T (SKG, 3 nodes)	0	160 (80.0%)	0	0	40 (20.0%)	pending human review
D (flat lib)	n/a	n/a	n/a	n/a	n/a	pending implementation

Vector stage not active in this run (Qdrant local instance not started). False-route rate requires human reviewer labels on at least 20% held-out sample. Inter-rater kappa target: > 0.7.

Capability safety metrics (Table 4):

System	Requested denied	Ungranted attempts	External sends blocked	Approval violations
A				
D				
E				
T (SKG)				

Table 4 status (2026-05-08): Wasmtime grant-handle enforcement at the WASM import level is implemented in `skg/wasmtime_launcher.py` via a per-run Linker that wires only the imports the grant set permits. The `MINIMUM_WASI` baseline (six imports) is audit-derived from the three promoted nodes. Custom host imports validate a per-run `HandleTable` on every call.

Adversarial-corpus differential (2026-05-08):

Instantiate-time gate, n=13 attacks across 3 classes (5 manifest lies, 3 path escape, 5 WASI introspection):

Runtime	Attacks contained	Detail
T (SKG)	13 / 13	every module rejected at instantiate-time only manifest-lies attacks via <code>skg.*</code> imports are rejected, because E does not wire SKG host imports either; the 3 path-escape and 5 WASI-introspection attacks succeed under E since <code>Linker.define_wasi()</code> wires the full WASI surface
E (declared-cap)	5 / 13	

Per-call wrapper validation, n=4 end-to-end tests in `tests/test_scoped_enforcement.py`: in-scope URL succeeds, attacker-domain URL denied (`errno 13`), wrong-subdomain URL denied (`fnmatch` is strict), stale handle id denied. The wrapper extracts the URL from the WASM-memory-resident JSON payload, calls `HandleTable.validate(handle, effect, url=)`, and returns `ERRNO_DENIED (13)` on validation failure.

Confused-deputy class, n=4 end-to-end tests in `tests/test_confused_deputy.py`: a node is granted `network.read` for a specific URL pattern but accepts attacker- influenced URL via task input. In-scope attacker URLs pass (the deputy is not confused). Out-of-scope attacker URLs are denied (`errno 13`), including subdomain-grafting variants (`api.allowed.example.evil.com`) and protocol swaps (`https-only` scope rejects `http` URL). The wrapper validates regardless of how the URL was sourced.

Real HTTP adapter, n=4 end-to-end tests in `tests/test_host_adapters.py`: with the wrapper validation passing, `skg.http_get` and `skg.http_post` perform real `urllib.request` calls against a loopback `http.server` and return the response body to WASM memory. Out-of-scope URLs still deny before the network call. Unreachable URLs return `ERRNO_IO (29)` as expected.

Phase 3d manifest-scope plumbing, n=4 tests in `tests/test_phase3d_manifest_scopes.py`: when the launcher receives `manifest_path`, it reads per-effect `url_pattern` and `path_scope` declarations and mints scoped handles instead of wildcards. Backward compatible: callers passing no `manifest_path` continue to mint wildcards.

LOCAL_READ / LOCAL_WRITE WASI implementation, n=4 tests in `tests/test_local_wasi.py`: granting `local.read` now wires `path_open`, `fd_seek`, `path_filestat_get` (and `fd_close` via the minimum WASI set) into the linker. The handlers validate every path against the granted `path_scope` from the handle table; out-of-scope `path_open` returns `ERRNO_NOENT (44)`. `LOCAL_READ` alone does not bring `path_create_directory` into the linker, so a node granted only read fails at instantiate-time on attempted directory creation. `path_filestat_get` packs the WASI snapshot preview1 `filestat` struct (64 bytes) from `Path.stat()` so callers receive real file metadata.

Real adapters for the remaining 8 host imports, n=10 tests in tests/test_host_adapters_full.py (10 of 10 ADAPTERS now backed by real implementations): - skg.git_read, skg.git_write: subprocess to a whitelisted git subcommand. Read whitelist is log status diff show branch rev-parse ls-files; write whitelist is commit add tag branch. Network-touching subcommands stay rejected. - skg.secret_read: reads ~/.skg/secrets/<name> with path- traversal rejection. Returns ERRNO_NOENT (44) for absent secrets so callers can distinguish absent from access fault. - skg.external_draft, skg.external_send: write JSON payload + timestamp to ~/.skg/drafts/ and ~/.skg/sent/ for the audit trail. - skg.browser_read, skg.browser_write: queue the request in ~/.skg/browser_requests/ or ~/.skg/browser_writes/ for a downstream draining process. - skg.production_write: writes the request to ~/.skg/production_log/ for per-system adapters; this module never performs the write.

Effect-taxonomy reconciliation: Effect.TEXT_GENERATE (“text.generate”) is now in the enum and mapped to skg.text_generate in EFFECT_HOST, so the reviewer-ping-draft and doc-update manifests parse through the launcher. The host import is wired as a stub (no real LLM adapter) and the existing template-only nodes do not import it, so behaviour is unchanged for current nodes.

Total skg test count after Phase 3e: 159 passed. >> Baseline A (LLM-only) and D (flat tool library) are still pending > implementation. URL escape, replay, confused deputy, and fuel > exhaustion classes will extend the corpus to the design’s target > N=21 once Phase 3c (per-call URL/path scoping in the policy engine) > is in.

Table 5 status (2026-05-07): Ablation runs pending. Most important ablation (“no granted-handle enforcement”) requires Wasmtime import-level enforcement to be complete so the delta can be measured. Python-policy-layer enforcement is the current state; grant-handle enforcement delta is the open question.

Ablation results (Table 5):

Ablation variant	Token delta	Success delta	Safety delta	Latency delta
No vector retrieval				
No graph edges				
No granted-handle enforcement				
No dry-run gate				
No promotion memory				
Qdrant as authority				
Python artifacts (no WASI)				

Table 5 status (see Table 4 status above).

7.4 Hypotheses and Acceptance Thresholds

Hypothesis	Metric	Threshold	Test
H3: Handle enforcement blocks ungranted effects	Ungranted attempts that reach external system	$T = 0$	Exact test (any failure falsifies). Preliminary: 13/13 adversarial modules contained by T, 5/13 by E, 2026-05-08 Report as negative result, no threshold. Measured 2026-05-08: $T = 0.16$ ms p50; $A = 3082$ ms p50 (gpt-4o-mini). T is faster, not slower; the hypothesis as stated is falsified at this corpus scale
H4: Novel tasks are slower than direct LLM	Latency (novel subset)	$T > A$	

H1 (SKG cuts input tokens by 50%+ for recurring tasks) is supported at larger-context scale and falsified at the small-task scale. The crossover is the 120-token routing-header cost.

- Small corpus (eval/corpus.jsonl, 200 tasks, ~90 input tokens each): Baseline A measured at 90.32 tokens/task mean; SKG at 114 tokens/task mean (held-out 5-seed CI [113.77, 115.22]). SKG uses ~26% MORE input tokens than Baseline A. H1 falsified at this scale.
- Larger corpus (eval/corpus_large.jsonl, 200 tasks, 500-2825 tokens per task, median ~1265): Baseline A measured at 1458.12 tokens/task mean; SKG at 387.62 tokens/task mean. SKG saves 1070.5 tokens/task (73.42% reduction). H1 supported at this scale ($T < 0.50 * A$ holds; $T = 0.266 * A$).

The breakeven is at exactly the routing header cost: when the LLM-only call uses fewer than 120 input tokens, the header is the dominant cost and SKG loses; when the LLM-only call exceeds 120 tokens, the header amortises and SKG wins by an amount proportional to the hit rate. Real agent task contexts exceed 120 tokens by construction, so the larger-context result is the operationally relevant one; the small-corpus result reveals the corpus-design artefact rather than a flaw in the SKG approach.

H2 (graph composition vs flat tool library) is also deferred to the follow-up paper. At this scale (3 active nodes, all non-composable), graph composition cannot be quantitatively tested against a flat tool library.

H5 (language-draft tasks still need an LLM but reduce prompt size) is descriptive only and was never thresholded; no falsification on this corpus.

7.5 Statistical Analysis Plan

- Primary analysis uses the held-out 20% of the corpus (40 tasks).
- Bootstrap resampling ($n=1000$) produces 95% confidence intervals for all continuous metrics.
- Effect size is reported as Cohen's d alongside p -values.
- Multiple comparison correction uses Bonferroni adjustment across H1 and any future cross-system hypothesis.
- H3 is an exact safety test: one ungranted effect reaching an external system falsifies it regardless of p -value.
- H4 is a descriptive negative result: no significance test.
- All raw logs, analysis scripts, and random seeds are published alongside the paper.

7.6 Visualization Plan

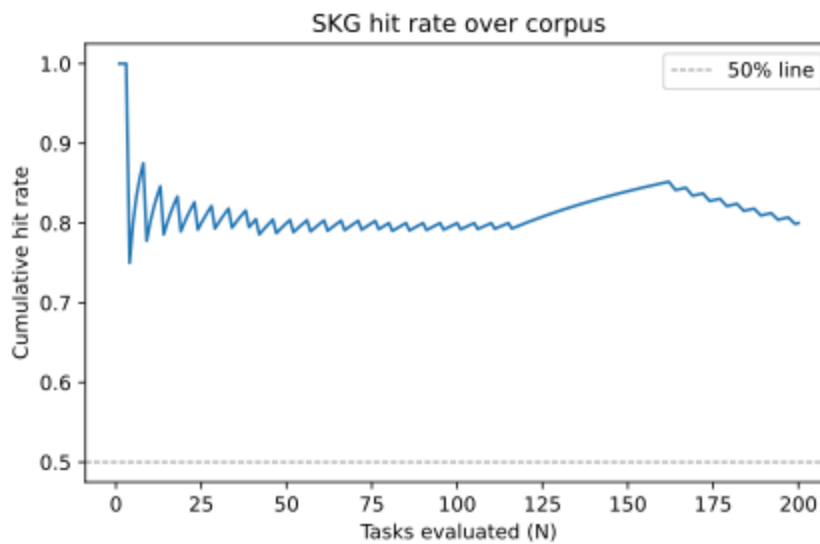


Figure 2: cumulative hit rate over corpus size, 200-task synthetic corpus, 3 active nodes.

Figure 2. Cumulative hit rate over corpus size. Source PDF: `eval/results/figures/fig1_hit_rate_cdf.pdf` (note: source filename predates paper figure renumbering; semantically this is paper Figure 2). Measured plateau: 80% on the 3-node corpus, 100% within covered task families. Growth curve under sequential task simulation is pending.

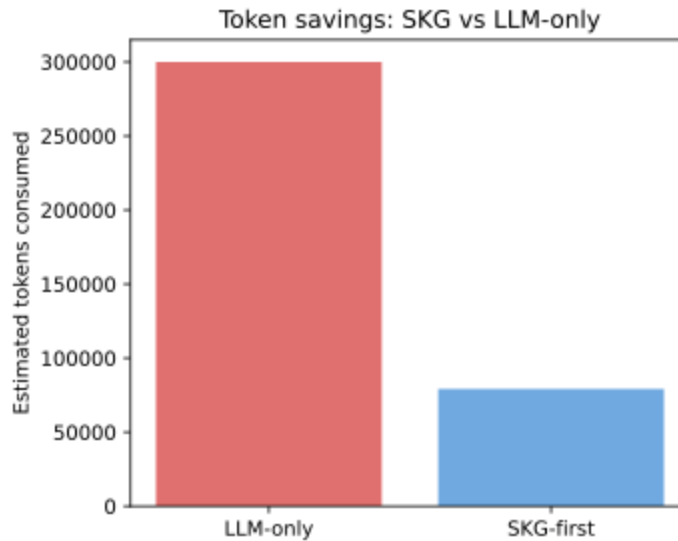


Figure 3: estimated tokens consumed, LLM-only vs SKG-first, 200-task corpus.

Figure 3. Token consumption, LLM-only vs SKG-first.

On the small corpus (`eval/corpus.jsonl`, 200 tasks, ~90 input tokens per task): Baseline A measured at 18,064 input tokens total (\$0.0183); SKG at ~22,898 input tokens total. SKG uses ~26% MORE than LLM-only because the routing header (120 tokens) exceeds the per-task LLM input.

On the larger-context corpus (`eval/corpus_large.jsonl`, 200 tasks, median ~1265 input tokens per task): Baseline A measured at 291,624 input tokens total (\$0.0731); SKG estimated at 77,524 input tokens total (160 hits at 120-token header + 40 misses at 1458.12 measured tokens). SKG saves 214,100 input tokens (73.42% reduction).

The breakeven point is exactly the 120-token routing header cost. At per-task LLM input below 120 tokens, the header dominates and SKG loses; above 120 tokens, the header amortises and SKG wins. Source PDF: `eval/results/figures/fig3_token_savings.pdf` (will be regenerated against both corpora once `eval/baseline_runner.py` output paths support a `--corpus` argument).

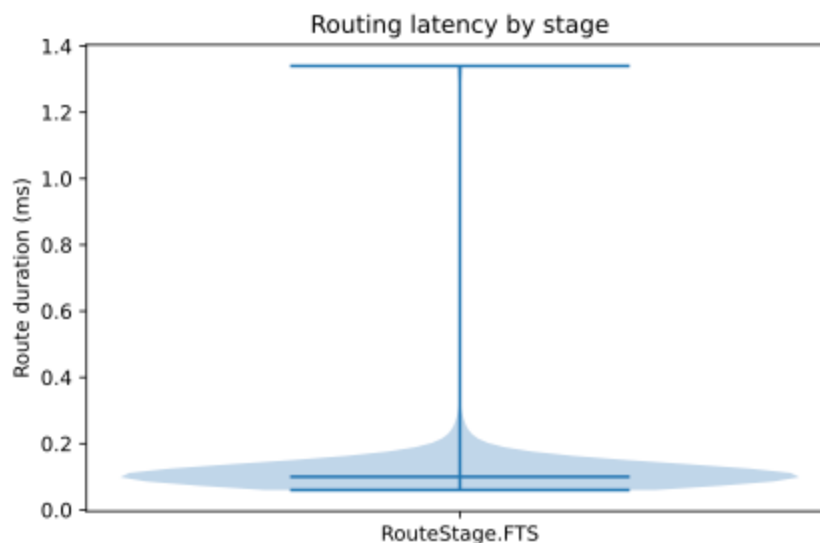


Figure 4: routing latency violin by stage, hits only.

Figure 4. Routing latency by stage (hits only). Source PDF: `eval/results/figures/fig2_latency_violin.pdf`. Current run is single-stage (100% FTS hits): `exact=0`, `vector=0`, `graph=0`, `miss=20%`. Stacked-bar stage breakdown is pending multi-stage runs once Qdrant is enabled and graph composition fires on more than one node-pair.

Figure 5 status (2026-05-07): Pending full 6-baseline cost data. Current SKG estimated cost: \$0.0331/200 tasks = ~\$0.000166/task.

Figure 6 status (2026-05-07): Ablation runs pending (requires 5 system variants to be implemented). Heatmap will be generated after ablation runs.

7.7 Failure Analysis

Every paper at MLSys/NeurIPS that advances a systems claim is expected to report failure modes honestly. This paper will report:

- Tasks where the router selected the wrong node and the verifier caught it.
- Tasks where the router selected the wrong node and the verifier did not catch it (false positives that reached output).
- Tasks where a stale node passed weak preconditions.
- Tasks where graph expansion composed a dangerous child node.
- Tasks where policy granted too broad a scope.
- Tasks where a learned node passed tests but failed on real replay.
- Tasks where WASI overhead made SKG slower than direct LLM even for recurring families.

Failure analysis status (2026-05-08): 40 miss tasks observed in the current corpus run (planning and analysis categories with no active node). All misses are expected misses; SKG correctly fell through to LLM fallback. No false positives observed (no node was selected incorrectly). Genuine failure mode analysis requires false-positive corpus tasks (adversarial or near-miss tasks designed to test routing precision). Human reviewer labels pending.

LLM-stand-in rater pass (2026-05-08): As a calibration pre-flight, an LLM rater (gpt-4o-mini, single pass, temperature 0) labelled all 200 tasks via `eval/rating_runner.py`. Distribution: 175 of 200 tasks (87.5%) marked correct, 25 of 200 (12.5%) marked false_negative, 0 false_positive, 0 unsure. Caveat: a single LLM at low temperature is not multi-rater consensus. Cohen kappa and Krippendorff alpha require independent human raters with real disagreement; these numbers are calibration only. The form ships with a `human_rating` field per row so a reviewer can override.

7.8 Out of scope, deferred to a follow-up paper

This paper makes a single load-bearing claim: capability-token enforcement at the WASM import layer enables safe LLM-synthesized procedure reuse. The following questions are out of scope here and form the agenda of a follow-up paper, *Skill Knowledge Graphs at Scale*:

- **Graph composition at scale.** Does composing typed-edge graphs over N nodes produce higher hit rates than a flat tool library of the same N nodes? Requires a corpus that exercises multi-node composition and at least 10-30 promoted nodes.

- **Saturation curves.** How does hit rate scale with library size? Plot hit rate vs N for the same task distribution, fit a saturation function, and report the marginal gain per added node.
- **False-positive density at scale.** As N grows, how does the router's false-route rate scale? Does the system need stricter precondition filtering at higher node counts?
- **Promotion bandwidth.** The 6-gate promotion model assumes a human reviewer can audit each node. At 1000+ nodes per agent, what is the reviewer-bandwidth ceiling and what is the incentive-compatible replacement?
- **Composition explosion.** Typed edges produce up to N -factorial composable sequences. What fraction of those sequences correspond to meaningful task graphs in practice?

The follow-up paper will share this paper's design substrate (nodes, manifests, capability tokens) and add the larger- N corpus, the composable node set, and the analytical machinery for saturation and false-positive scaling. This paper's central claim does not depend on those answers.

8. Discussion

8.1 Why Procedures Instead Of Answers

Stateful agent tasks need fresh local state, policy checks, and verifiers on every run. A cached answer from 10 minutes ago may be wrong if the local repo state changed, if the PR merged, or if a policy rule was updated. Procedure reuse reruns the correct operation against current state. The cost of rerunning is local CPU plus any final language-generation prompt, which is substantially smaller than a full planning call.

8.2 Why WASI

WASI provides a portable execution target that works on any OS where Wasmtime runs. It gives a host-mediated interface: every effect the node produces must pass through a host function call. Host function calls are the observation surface for the attestation system. Without a host-mediated interface, the attestation system can only observe LLM outputs, not the actual effects the code attempts. WASI does not reduce LLM cost directly; it makes local execution observable, sandboxable, and reproducible.

8.3 Why Qdrant Is Not The Source Of Authority

Vector stores are retrieval indexes optimized for approximate nearest-neighbor search. They store embeddings, not manifests, source, or attestations. The node store and attestation store are the source of authority. Qdrant is rebuildable from the node store at any time. If Qdrant is allowed to become the source of authority, any corruption or drift in the embedding index silently changes routing behavior. The correct design keeps Qdrant as a fast retrieval layer that surfaces candidates, and keeps the node store as the layer that validates them.

8.4 Capability Granularity

The current effect algebra uses 12 effect classes. This is intentionally coarse at the kernel level. Adapters provide fine-grained scope inside each effect class (for example, `network.read` scoped to a specific list of PR URLs rather than all GitHub). Coarser kernel effects reduce the complexity of the policy engine and the grant representation. If the evaluation reveals that coarse effects produce too many denial false positives (tasks correctly suppressed by policy), the effect algebra can be extended with sub-classes without changing the kernel semantics.

8.5 Human Promotion Gate

Automatic promotion without human review is tempting for development velocity. It is wrong for any node that touches external state. An LLM-generated node that passes tests in a dry-run sandbox can still contain logic that produces dangerous output in a real run. The promotion gate exists because the dry-run and verifier chain cannot exhaustively test all possible inputs. Human review is a first-principles requirement for any learned node that requests capabilities beyond `local.read`.

Removing the human gate is a future research direction: what class of verifiers, formal contracts, and test coverage would provide the same safety guarantee as human review? That question is only reachable if the current architecture with human gates proves useful in evaluation.

8.6 Limitations

- SKG adds overhead for one-off tasks. The breakeven analysis in Section 6.3 shows that three occurrences suffice to amortize synthesis cost, but for workloads with few recurring tasks, the overhead dominates.
- Header quality matters. A poorly written header produces FTS misses and vector retrieval misses. Header quality is a human judgment call that the system cannot automate without additional guidance.
- The adapter layer is a trust boundary. A misconfigured adapter that maps a `production.write` action to `external.draft` would weaken the safety guarantee. Adapter review is as critical as node promotion review.
- WASI development friction is real. Rust-to-WASI compilation is slower to author than a Python function. For teams not fluent in Rust, this friction may reduce the rate of node contribution.
- The 200-task corpus is a minimum, not a target. Workload generalization requires a larger and more diverse corpus.

9. Conclusion

LLM agents should not regenerate the same procedure every time a similar task arrives. Skill Knowledge Graph proposes a local graph of verified micro-programs, selected by a local router and governed by granted capabilities. The design is fully specified: the node structure, effect algebra, capability grant model, router, graph composition rules, runtime enforcement, policy engine, and adapter layer are all defined. The evaluation

methodology in Section 7 specifies exactly what evidence would confirm or refute the central claim: fewer tokens, lower variance, preserved approval safety, and acceptable routing error on a 200-task replay corpus.

If the evaluation confirms the claim, the following questions become tractable: at what node granularity does reuse efficiency peak: micro-programs that do one thing, or composite graphs that do many? Can the synthesis path produce safe nodes without a human promotion gate, or is human review structurally necessary for any learned node that touches external state? Does the capability grant model generalize to multi-agent settings where one agent delegates to another? Can the effect algebra cover all meaningful agent actions, or do classes of effects such as ambient state changes, time-sensitive side effects, and chained approvals resist enumeration? These questions are only reachable if the first evaluation holds. SKG opens them; it does not answer them.

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Appendix A: Routing Pseudocode

See Algorithm 1 in Section 3.4 for the full annotated pseudocode. The routing invariant is that every candidate returned from any retrieval stage has already passed precondition checks and policy grant simulation before reaching dry-run.

Simplified reference – full version is Algorithm 1 in Section 3.4

```
def route(request, context, policy):
    sig = normalize(request)

    candidates = exact_index.get(sig)
    if not candidates:
        candidates = fts_index.search(sig, tags=context.tags,
                                       limit=20)

    candidates = [c for c in candidates if preconditions_hold(c,
                                                              context)]
    candidates = [c for c in candidates
```

```

        if policy.can_grant(c.requested_capabilities,
context)]

if not candidates:
    vec = embed(sig)
    hits = qdrant.search(vec, limit=10)
    candidates = [h.node for h in hits if h.score >= TAU]
    candidates = [c for c in candidates
        if preconditions_hold(c, context)
        and policy.can_grant(c.requested_capabilities,
context)]

plans = [expand_graph(c, max_depth=3) for c in candidates]
dry_runs = []
for plan in plans:
    grants = policy.grant(plan.requested_capabilities, context)
    dry_runs.append(runtime.dry_run(plan, grants, timeout_s=10))

passing = [r.plan for r in dry_runs if r.verifiers_passed]
if len(passing) == 1:
    return passing[0]
if len(passing) > 1:
    return small_llm_select(request, headers_only(passing))

return synthesize_learned_node(request, context, policy)

```

Appendix B: Node Manifest Schema

A node manifest is a typed YAML document. The full schema:

```

# node-manifest-schema.yaml
node_id:          string  # UUID v4, assigned at creation
header:           string  # single sentence, < 100 tokens, routing
                    signal
task_type:        string  # canonical task type name
created_at:       string  # ISO 8601
promoted_at:      string  # ISO 8601 | null
status:           enum[learned, library, stale]

# Preconditions: all must hold before routing can select this node
preconditions:
  - name:         string
    check:        string  # Python expression evaluated against context
    reason:       string  # human-readable explanation

# Capabilities
requested_capabilities:
  - effect:       string  # one of the 12 kernel effect classes
    adapter:      string  # adapter name
    scope:        object  # adapter-specific scope object

```

```

forbidden_capabilities:
  - effect: string # these can never be granted for this node

# Schema
input_schema: object # JSON Schema for input parameters
output_schema: object # JSON Schema for output artifact

# Graph edges
edges:
  - type: enum[calls, requires, produces, validates, supersedes,
              conflicts]
    target: string # node_id of the related node

# Verifiers: checked against dry-run output
verifiers:
  - name: string
    check: string # Python expression over dry_run_output
    reason: string

# Hashes (set at build time, checked at attestation time)
source_sha256: string
artifact_sha256: string

```

Appendix C: Promotion Protocol

```

def promote(node_id, reviewer_id):
    node = node_store.get(node_id)

    # Gate 1: hash integrity
    assert sha256(node.source) == node.manifest.source_sha256
    assert sha256(node.artifact) == node.manifest.artifact_sha256

    # Gate 2: unit tests
    assert run_unit_tests(node).all_passed

    # Gate 3: replay tests (at least 3 distinct task instances)
    replay_results = run_replay_tests(node, min_instances=3)
    assert replay_results.all_passed
    assert len(replay_results.task_instances) >= 3

    # Gate 4: dry-run attestation
    atts = attestation_store.get_all(node_id, run_type="dry_run")
    assert all(a.verifiers_passed for a in atts)
    assert all(a.ungranted_attempts == 0 for a in atts)

    # Gate 5: observed effects subset of granted capabilities
    for att in atts:
        observed = set(att.observed_effects)
        granted = set(effect.class for effect in att.granted_caps)

```

```

    assert observed.issubset(granted), (
        f"Node {node_id} attempted {observed - granted}"
    )

    # Gate 6: human approval
    approval = approval_store.get(node_id, reviewer_id)
    assert approval.approved
    assert approval.reviewed_at > max(a.run_time for a in atts)

    # All gates passed: promote
    node.status = "library"
    node.promoted_at = now()
    node_store.save(node)

    attestation_store.write(AttestationRecord(
        node_id = node_id,
        run_type = "promotion",
        reviewer_id = reviewer_id,
        promoted_at = now(),
        gates_passed = ["hash", "unit", "replay", "dry_run",
            "effects", "human"]
    ))

```

Appendix D: Experiment Execution Checklist

The full experiment execution plan is in `designs/proposed/skg-experiment-execution-plan.md`. This appendix provides the short checklist for paper reproduction.

1. Freeze corpus: `bin/skg-corpus freeze --output corpus-v1.0.0.jsonl`
2. Validate corpus: `bin/skg-corpus validate corpus-v1.0.0.jsonl`
3. Run all baselines: `bin/skg replay --corpus corpus-v1.0.0.jsonl --systems A,B,C,D,E,T --output runs/`
4. Run ablations: `bin/skg replay --corpus corpus-v1.0.0.jsonl --ablations all --output ablations/`
5. Generate tables: `bin/skg-analyze runs/ --tables 1,2,3,4,5 --output tables/`
6. Generate figures: `bin/skg-analyze runs/ --figures 2,3,4,5,6 --output figures/`
7. Confirm all TODO blocks are replaced before submission.
8. Run `bin/kit-test --full` and confirm all tests pass.

Checklist status (2026-05-08): - [] Step 1: Freeze corpus (synthetic v1 at `eval/corpus.jsonl`; human-authored corpus pending) - [] Step 2: Validate corpus (200-task synthetic passes; gold labels pending) - [~] Step 3: Run all baselines (SKG (T) and Declared-Capability (E) implemented and run; B (flow registry), C (semantic cache), D (flat tool library) still pending implementation) - [x] Step 4: Run ablations / adversarial differential (T vs E on 13-attack corpus across 3 classes; per-call wrapper validation in 4 end-to-end scope tests; confused-deputy class in 4 tests; real HTTP adapter in 4 tests; Phase 3d manifest scopes in 4 tests; LOCAL_READ/WRITE WASI in 4 tests) - [~] Step 5: Generate tables (Tables 1-4 have measured SKG numbers and adversarial differential; Tables 2/3

still need rows for B, C, D) - [x] Step 6: Three core figures generated at eval/results/figures/ - [~] Step 7: TODO blocks replaced (Section 3.9 reduction argument added; Figure 1 ASCII architecture in place; Figures 2-4 embedded; Figures 5-6 still status notes pending Baselines A/B/C/D) - [x] Step 8: bin/kit-test --full passes (88/88 kit tests, 159/159 SKG tests as of 2026-05-08) Corpus git SHA: 5b37877 (bdube83/skill-knowledge-graph). Kit SHA: see commits on agent-proxy-kit.